Solution:

Step 1 : Hypotheses

Null Hypotheses: $H_{01}: \mu_{M_1} = \mu_{M_2} = \mu_{M_3}$ (for treatments)

That is, there is no significant difference among the three programs in their mean sales.

$$H_{02}: \mu_{S_1} = \mu_{S_2} = \mu_{S_3} = \mu_{S_4} \text{ (for blocks)}$$

Alternative Hypotheses:

H_{11} : At least one average is different from the other, among the three programs.

H_{12} : At least one average is different from the other, among the four salesmen.

Step 2 : Data

Salesmen	Methods		
	A	B	C
1	4	6	2
2	6	10	6
3	5	7	4
4	7	5	4

Step 3 : Level of significance $\alpha = 5\%$

Step 4 : Test Statistic

$$F_{0t}(\text{treatment}) = \frac{MST}{MSE}$$

$$F_{0b}(\text{block}) = \frac{MSB}{MSE}$$

Step-5 : Calculation of the Test Statistic

	Methods			Total $x_{i.}$	$x_{i.}^2$
	A	B	C		
1	4	6	2	12	144
2	6	10	6	22	484
3	5	7	4	16	256
4	7	5	4	16	256
x_i	22	28	16	66	1140
$x_{i.}^2$	484	784	256	1524	

Squares

16	36	4
36	100	36
25	49	16
49	25	16
		$\sum\sum x_{ij}^2 = 408$

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